PRATHIK KUMAR

ELECTRONICS & COMMUNICATION ENGINEERING GRADUATE (2027)

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Electronics & Communication Engineering student with hands-on experience across embedded systems, IoT product development, FPGA physical design, and full-stack application development. Proficient in Embedded C, C++, and Python, with practical exposure to sensor integration, communication protocols (UART, SPI, I2C, MQTT), and FPGA timing optimization using Vivado and RapidWright. Strong analytical foundation shaped by translating system-level requirements into working hardware-software solutions across multiple internships and academic projects.

1. EDUCATION DETAILS

Course Name	Institute Name	University Name	Year of Passing	Score Details (%/CGPA)	Course Type
B.Tech (ECE)	Anurag University	Anurag University	2027 (Expected)	8.5 CGPA	Full-time
Intermediate (12th)	Deeksha Junior College	Board of Intermediate Education, Telangana	2023	9.94 GPA	Full-time
10th	CAL Public School	Board of Secondary Education, Telangana	2021	94.4%	Full-time

2. PROJECTS UNDERTAKEN

ENVY — Smart Agriculture Environmental Monitoring System

- Co-developed a full-stack IoT platform using an ESP32 microcontroller interfaced with 9 environmental sensors to help farmers monitor field conditions.
- Worked on sensor interfacing, embedded firmware for data acquisition, and the hardware-to-cloud data pipeline (ESP32 → Supabase/PostgreSQL → Flutter mobile app), with an NVIDIA NIM-based layer for data-driven recommendations.
- Delivered a working end-to-end prototype validated across all 9 sensor channels; co-authored an IEEE research paper detailing the system architecture and sensor methodology.
- Project has drawn interest for a patent filing; team is now planning a move to industrial-grade components.

Environmental Intelligence Network — Multi-Hazard Early-Warning IoT System (SIH, Qualcomm Problem Statement)

- Designed and built a distributed environmental hazard-detection system on ESP32, integrating 9+ sensors (gas, water quality, soil, seismic, weather) with four independent on-device 1D-CNN models (TensorFlow Lite Micro, INT8-quantized) for real-time flood, fire, air-quality, and landslide risk inference.
- Built a dual-layer safety architecture combining deterministic threshold rules with CNN-based anomaly detection, enabling local decision-making without cloud dependency and immediate SMS alerts via GSM for zero-latency community warnings.
- Developed a cloud backend (FastAPI + Supabase) with a React/Leaflet dashboard performing inverse-distance-weighted spatial interpolation for regional risk heatmaps.

FPGA Optimization Engine (PraxOptimizer) — FPL'26 FPGA Backend Optimization Competition

- Built a deterministic FPGA physical-optimization engine on Vivado and RapidWright using empirically-validated hardcoded pblock ranges, multi-pass phys_opt_design strategies, and stagnation detection.
- Designed the optimization strategy logic, debugged the Vivado/RapidWright integration, and tuned runtime performance across multiple tool versions.
- Developed a deterministic UCB-style multi-armed bandit scheduler and a 17-recipe Fmax optimization library spanning Tier 1–4 benchmarks, targeting UltraScale+ devices; entered the tool into the FPL'26 contest, iteratively improving it against competition benchmarks.

Stock Market Prediction Using Z-Transform

- Signals and Systems course project applying Z-transform analysis in MATLAB to model and predict stock market price movements.

TJASF — The Journal of Advanced Scientific Frontiers (Live Website)

- Designed the UI/UX for TJASF's website, a live journal platform co-founded with a faculty member, alongside branding and template materials including logo variants, an IEEE-style submission template, and a journal launch package.

3. STRONG SUBJECTS

- Signals and Systems
- Digital Logic Design & Computer Architecture
- Embedded Systems & Microcontrollers
- Data Structures & Algorithms
- Object-Oriented Programming
- VLSI Design Fundamentals

4. CERTIFICATIONS

- **ServiceNow Virtual Internship Program**
- **Google Cloud AI & Machine Learning Specialization**
- **Gemini AI & Responsible AI**
- **Introduction to Cybersecurity**
- **Cybersecurity Essentials**
- **Google UX Design Professional Certificate**
- **UI/UX Design with Figma (Udemy)**

5. INTERNSHIP EXPERIENCE

Intern, FlyRank **July 2026 – Aug 2026 | Remote**

- Collaborated with product and engineering teams on technical documentation and cross-functional workflows.
- Translated system requirements into developer-ready specifications.

Junior Design Intern, SuPrazo Technologies **Dec 2025 – May 2026 | Remote / Hyderabad**

- Collaborated with cross-functional engineering and design stakeholders on component development and technical reviews.
- Participated in iterative feedback cycles for internal and client-facing products.

Virtual Internship, ServiceNow **Virtual**

- Completed a hands-on virtual internship covering application development and workflow automation on the Now Platform.
- Practiced scripting and configuration to build and customize workflow-driven applications; gained exposure to enterprise software development practices, including requirements analysis and platform-based app building.

ADDITIONAL INFORMATION

- **Leadership:** President – Cygnus Club (led 100+ members across design, AI, and product workshops; conducted technical sessions and tool development) | Design Lead – Robotics Club & IETE Chapter (built design systems & technical branding, mentored junior members).
- **Languages:** English, Hindi, Telugu.